

## ***B.Tech. Degree II Semester Regular and I Semester Supplementary Examination May 2016***

### **CE/EE 1203 A ENGINEERING GRAPHICS**

(2015 Scheme)

Time : 3 Hours

Maximum Marks : 60

(5 × 12 = 60)

- I. Construct a scale of RF 1/25, to show metres and decimetres and by a vernier to read centimetres and long enough to measure up to 4 metres. Show a distance of 2.48 metres and 3.01 metres on it.

**OR**

- II. A fish pond of elliptical shape is to be inscribed inside a rectangular plot of size 100 m × 50 m. Draw the boundaries line of the fish pond (write scale used).

- III. A room is 4.8 m × 4.2 m × 3.6 m high. Determine graphically, the distance between a top corner and bottom corner diagonally opposite to it.

**OR**

- IV. A square plate of side 40 mm has its surface perpendicular to both HP and VP one of the sides of the plate is inclined at 30° to HP. Draw its projections.

- V. Draw the projections of a cube of 35 mm edge resting on HP on one of its corners with a solid diagonal perpendicular to the VP.

**OR**

- VI. A pentagonal pyramid of base side 40 mm and axis length 75 mm is resting on HP on its base with one of its base side parallel to VP. It is cut by a plane inclined at 35° to HP and perpendicular to VP, bisecting the axis. Draw sectional top view and true shape of the section.

- VII. A hexagonal pyramid of base side 30 mm and height 60 mm rests vertically on HP with one of the base sides parallel to VP. It is cut by a section plane inclined at 45° to the HP, perpendicular to the VP and passing a point through the axis 10 mm below the apex of the pyramid. Draw the development of the bottom part of the pyramid.

**OR**

- VIII. A vertical cylinder of base diameter 70 mm is penetrated by a horizontal cylinder of 70 mm diameter, the axis of which is 10 mm in front of the axis of vertical cylinder. Draw the projections showing the curves of intersection when the axis of the horizontal cylinder is parallel to VP and HP.

- IX. Draw the isometric projection of a hexagonal prism of base side 30 mm and axis length 60 mm rests on HP on one of its rectangular faces with its axis perpendicular to VP.

**OR**

- X. A solid is in the form of a square prism of side of base 20 mm up to a height of 35 mm and thereafter tapers in to the frustum of a square pyramid, whose top surface is a square of side 10 mm. The total height of solid is 50 mm. Draw the solid in perspective, given that a side of base of the solid is resting on the ground parallel to the picture plane. The end of the side nearest to the eye is 20 mm to the right of the eye and 15 mm behind the picture plane. The eye is 70 mm from the picture plane and 60 mm above the ground.

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